

Analysis of Consumer Price Index Trends [CPI] in India (2013–2024)

➤ Project Overview:

This project analyzes the Consumer Price Index (CPI) trends in India from 2013 to 2024. The objective is to understand inflation patterns across different consumer categories such as Food, Fuel, Housing, Health, and Education. The analysis helps in identifying long-term price movements and sector-wise inflation behavior.

➤ Objectives:

- Analyze CPI trends across India (2013–2024)
- Identify inflation patterns across sectors
- Compare category-wise price behavior
- Perform statistical and visual analysis
- Extract meaningful economic insights

➤ Dataset Information:

- Source : Dataset collected from Indian Government Portal
- Location : Rural/Urban (Pan India)
- Timeline : January 2013 to January 2024 ; Published On: 05/06/2024
- Domain : Statistical [Ministry of Statistics and Programme Implementation]
- Link :
<https://www.data.gov.in/resource/all-india-consumer-price-index-ruralurban-upto-january-2024>

➤ **Key Features:**

- Year
- Month
- Food & Beverages
- Fuel & Light
- Housing
- Health
- Education
- General Index

➤ **Tools & Technologies Used**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

➤ **Data Preprocessing**

- Handled missing values using median and mode
- Removed duplicate records
- Standardized column names
- Converted data types appropriately
- Treated outliers using IQR method
- Created new time-based features (Year-Month)

➤ Exploratory Data Analysis (EDA)

✓ Univariate Analysis

- Distribution of CPI values using histograms and boxplots
- Frequency analysis of categorical variables

✓ Bivariate Analysis

- Year-wise CPI trends
- Relationship between Food & Fuel prices
- Sector-wise comparisons

✓ Multivariate Analysis

- Correlation heatmap of CPI categories
- Pairwise relationships between sectors
- Multi-line trend comparisons

✓ Statistical Analysis

- Mean, Median, Mode analysis
- Variance and Standard Deviation
- Skewness and Kurtosis
- Distribution behavior across categories

➤ Key Insights:

- Food, Fuel, and Housing are the major drivers of inflation
- CPI shows a consistent upward trend over time
- Fuel prices show high volatility compared to other sectors
- Rural and Urban inflation trends move almost similarly
- Essential goods show steady long-term price increases

➤ Visualizations:

The project includes:

- Histograms
- Boxplots
- KDE plots
- Scatter plots
- Line charts
- Heatmaps
- Bar charts

➤ Future Enhancements:

- Use larger and more recent datasets for better accuracy
- Build automated dashboards (Power BI / Tableau)
- Apply time-series forecasting models (ARIMA, Prophet)
- Include macroeconomic indicators for deeper insights
- Develop real-time CPI tracking dashboard

➤ Conclusion:

This project provides a clear understanding of inflation trends in India over a decade. It highlights how essential commodities contribute to rising consumer prices and helps in understanding economic behavior across sectors.

***** The End *****